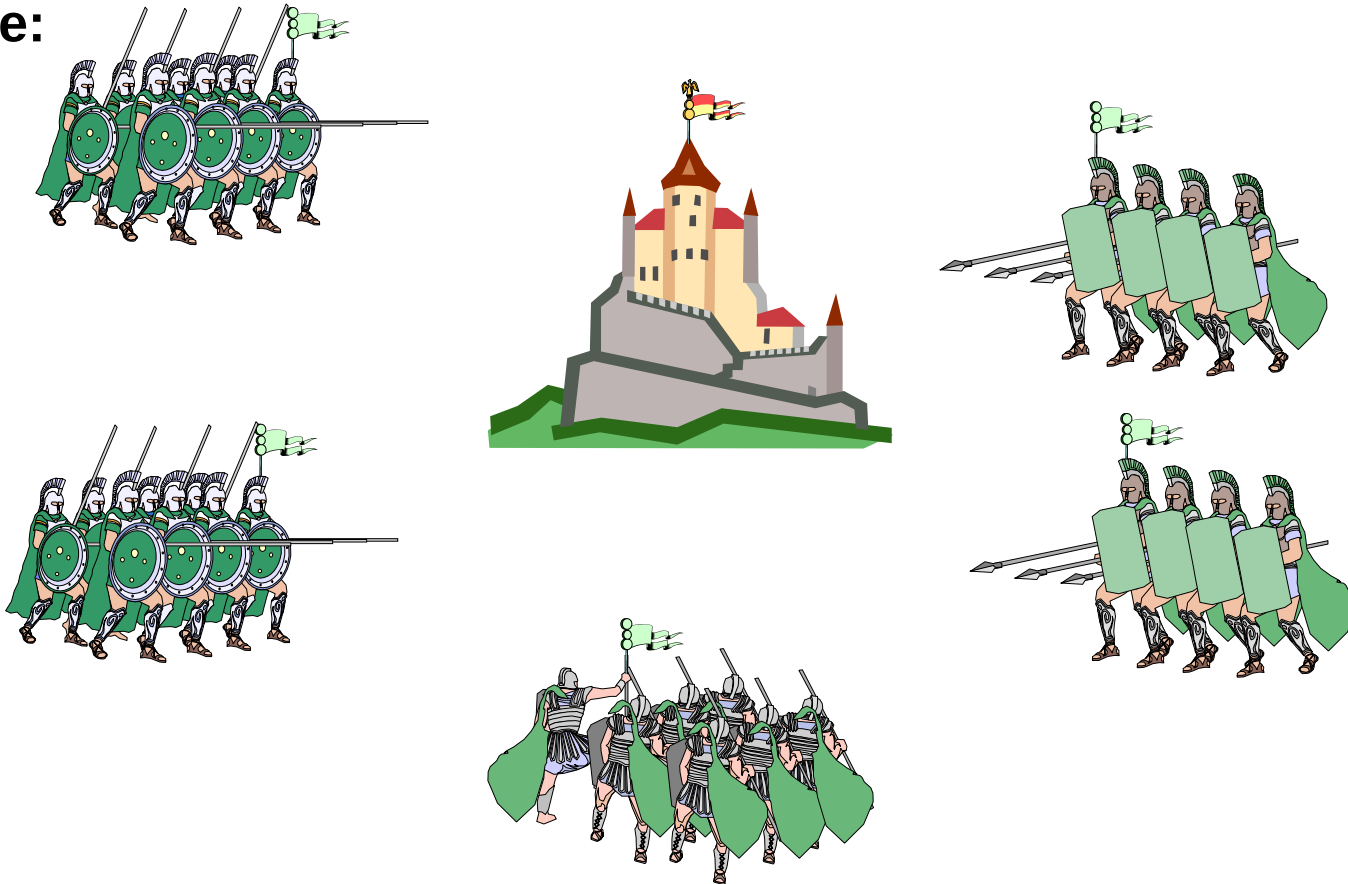


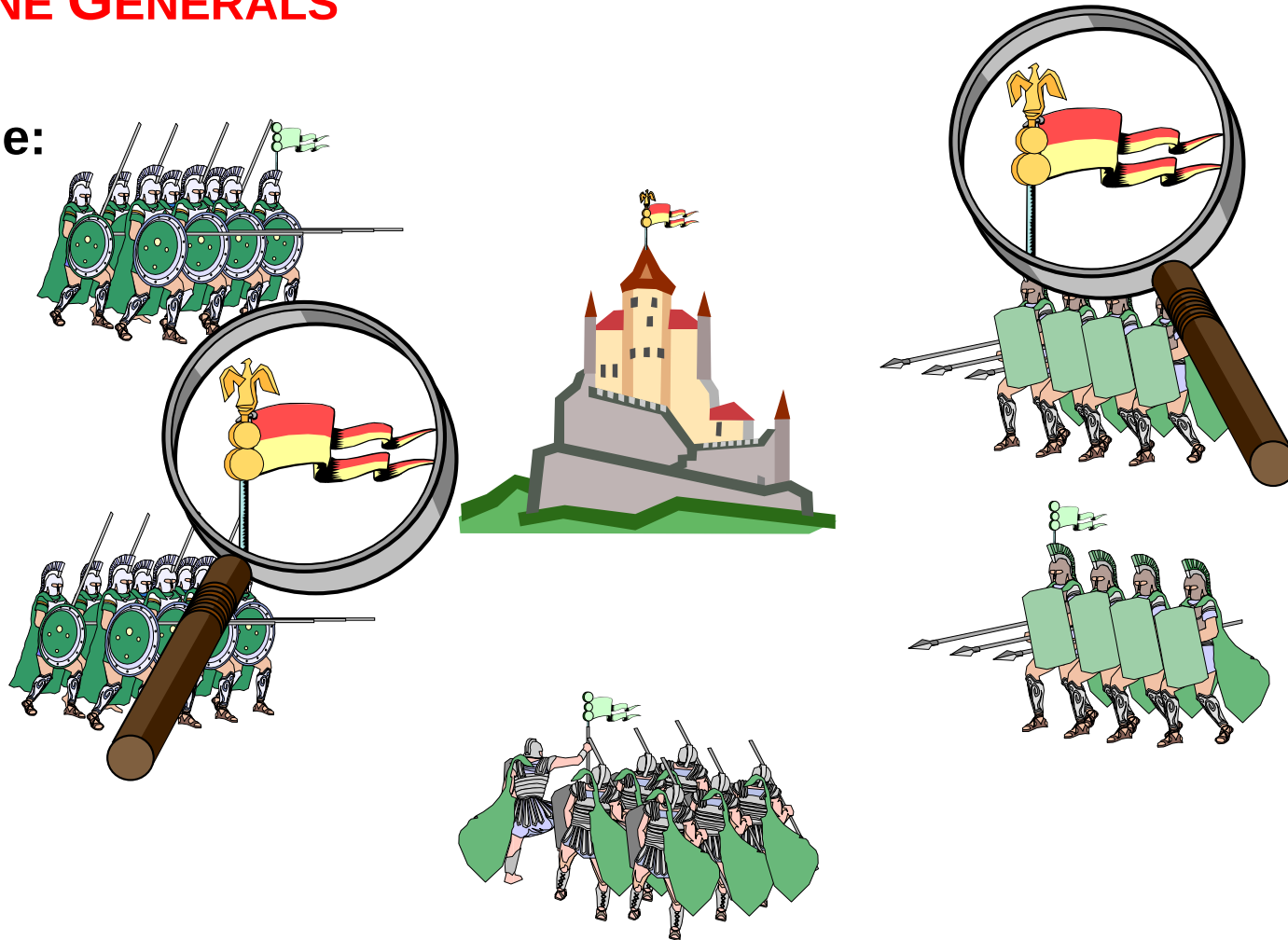
COORDINATED ATTACK PROBLEM

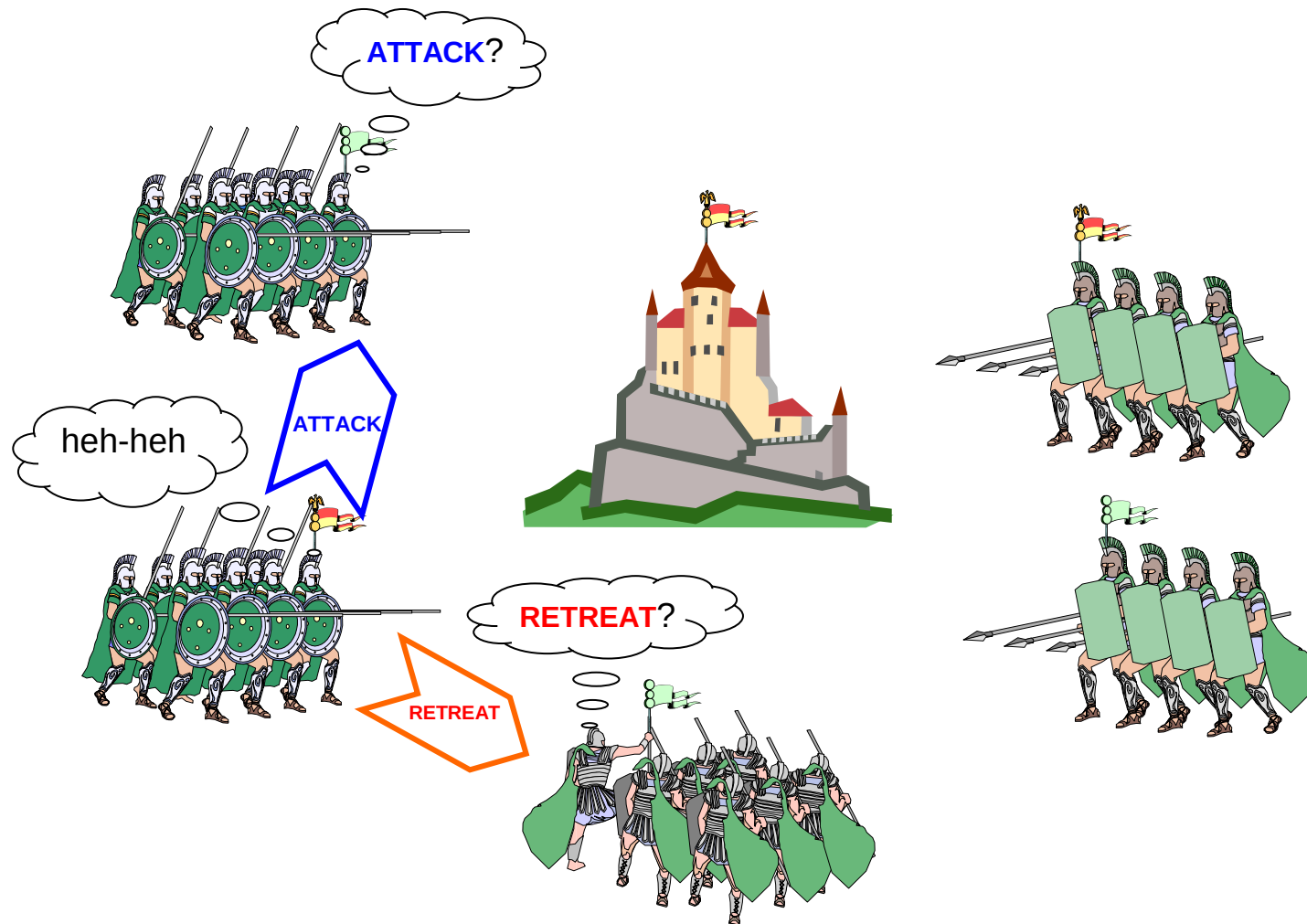
A fable:



BYZANTINE GENERALS

A fable:







Ad-hoc protocol:

failure-free
environment

- each general G_i makes a decision v_i and broadcasts it
- generals reach agreement based on the total of collected opinions:
 - if most of generals independently decide **ATTACK**, then attack
 - if most of generals independently decide **RETREAT**, then retreat
 - if they are divided, then ... hmm ... well, uh ... hmm ... who cares?
- it may possibly require several rounds

Byzantine failures:

- what's the problem? ... traitors !
- if you receive v_j from G_j , you can't trust him
- you must be careful, because you don't know who to trust
- it is intuitively obvious that unless a majority of nodes are honest, nothing useful can be done

Failures

Failure types:

- crash failures
 - omission failures
 - Byzantine (malicious) failures
- } fail-stop model

Omissions in Byzantine Generals model:

Since a faulty general (*traitor*) can refuse to send a message, a nonfaulty general may never receive the expected message. In such situation we assume that the nonfaulty general (*loyal*) simply chooses an arbitrary value and acts as if the expected message has been received.

Obviously, we require that such omissions can be detected by the respective receiver. In synchronous systems, where the duration of each round is known, this detection is simple – all expected messages not received by the end of a round were not sent (*omitted*).

BYZANTINE AGREEMENT (BA)

A commanding general G_0 sends an order to $N-1$ lieutenant generals and:

Agreement: All loyal lieutenants obey the same order.

Validity: If the commander is loyal, then every loyal lieutenant obeys the order he sends.

for the moment we skip

Termination: All loyal lieutenants eventually take an order.



Note that if G_0 is loyal, then *Agreement* follows directly from *Validity*.



IMPOSSIBILITY OF BYZANTINE AGREEMENT

Loyalty is important:

Claim: No solution for fewer than $3f + 1$ generals can cope with f traitors.

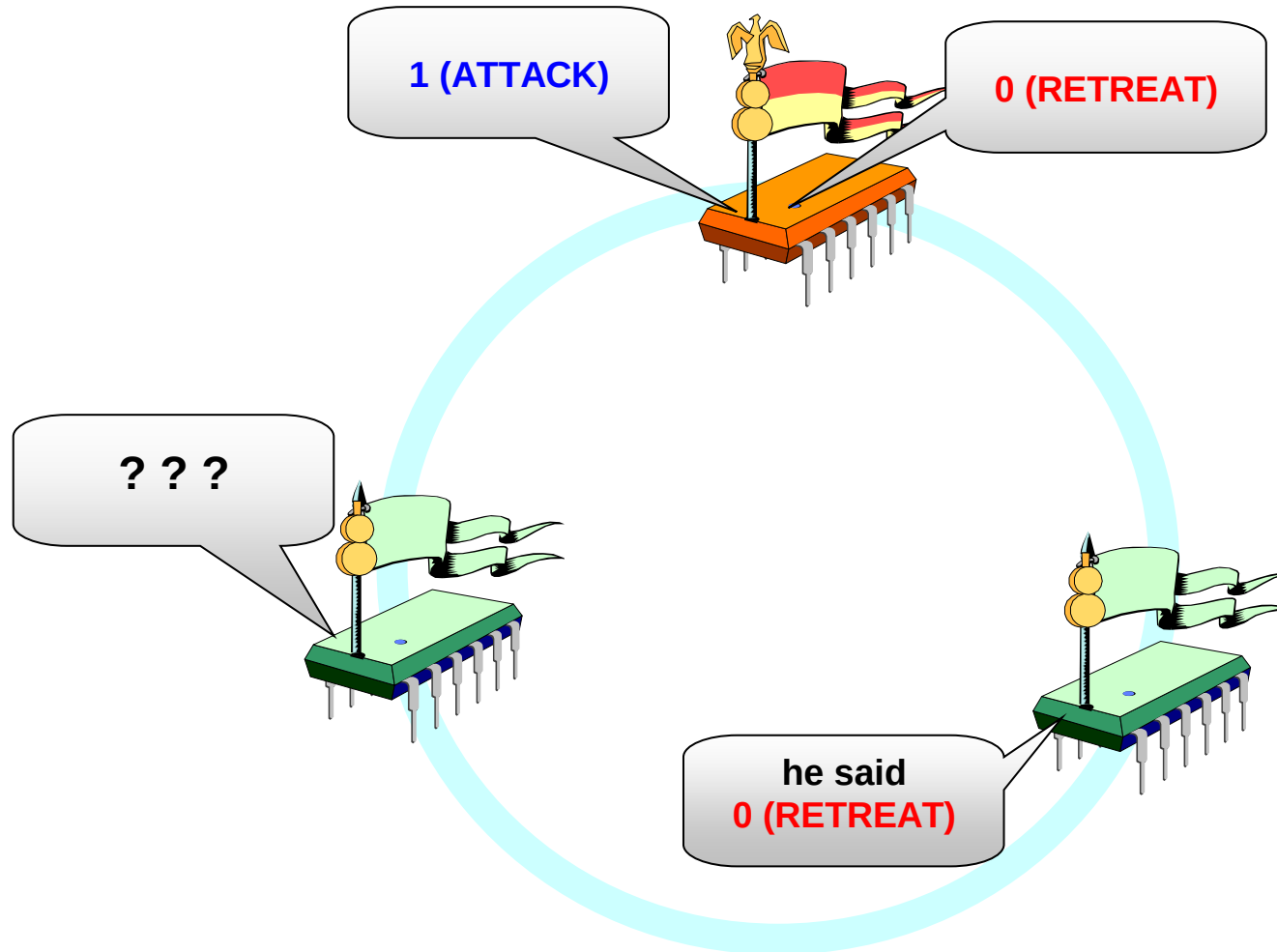
(i.e. the solution is impossible if $f \geq \frac{1}{3} N$ generals are traitors)

First we will show that no solution for 3 generals can handle a single traitor.

IMPOSSIBILITY OF BYZANTINE AGREEMENT

(2)

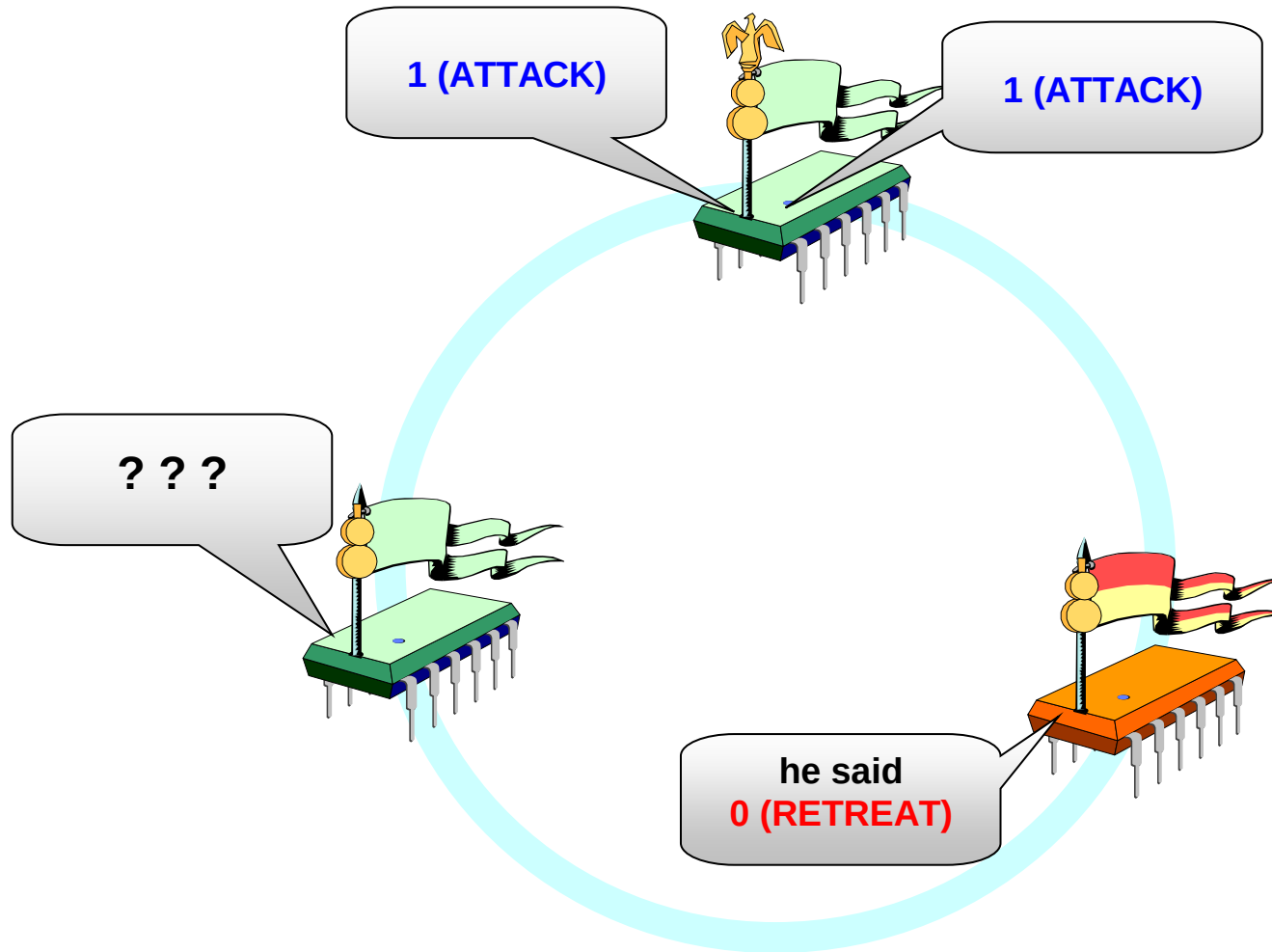
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IMPOSSIBILITY OF BYZANTINE AGREEMENT

(3)

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INFORMATYKI





Impossibility Result:

There is no f -resilient Byzantine Agreement algorithm in $\mathcal{S}^{\text{Sync}}$ for $f \geq \frac{1}{3}N$

Proof strategy:

- Assume we have an algorithm A_f for $3f$ generals with $f > 1$ traitors.
- We construct a 3-general 1-traitor algorithm A_1 using the following transformation:

- each general “simulates” f generals in A_f
- 1 traitor “simulates” f traitors
- 2 loyal generals “simulate” $2f$ loyal generals
- both *Agreement* and *Validity* are satisfied

- But this is impossible, so our assumption is wrong! ■



APPROXIMATE AGREEMENT (AA)

Maybe the difficulty is requiring exact agreement?

New problem:

G_0 sends an attack **TIME** to $N-1$ lieutenant generals:

Agreement: All loyal lieutenants attack within 10 minutes of each other.

Validity: If the commander is loyal, then every loyal lieutenant attacks within 10 minutes of the commander's order.

Question:

Is this agreement problem any easier?



AA Impossibility Result:

There is no f -resilient Approximate Agreement algorithm in $\mathcal{S}^{\text{Sync}}$ for $f \geq \frac{1}{3}N$

Proof strategy:

- Assume we have a 3-general 1-traitor Approximate Agreement algorithm.
- We will transform it into a 3-general 1-traitor Byzantine Agreement algorithm →



Transformation to Byzantine Agreement

Suppose G_0 sends:

- 1:00 to mean **ATTACK**
- 2:00 to mean **RETREAT**

Each lieutenant does:

Phase 1

- run the Approximate Agreement protocol
- if the agreed time is before 1:10, then decide to **ATTACK**
- if the agreed time is after 1:50, then decide to **RETREAT**

Phase 2

- if you don't come to any decision
- ask the other lieutenant "*Have you decided?*"
- if so do the same thing
- if not → **RETREAT**



Claim: If the Approximate Agreement protocol works, so does the Byzantine Agreement protocol.

Proof:

- If G_0 is loyal, then *AA Validity* ensures that the loyal lieutenant gets the right attack time, so *BA Validity* is satisfied (and so is *Agreement*).
- If G_0 is a traitor, both lieutenants are loyal. *AA Agreement* ensures they cannot make contradictory decisions in Phase 1.
- If one decides in Phase 1, the other agrees no later than in Phase 2.
- If neither decides in Phase 1, they both retreat in Phase 2.

Thus Approximate Agreement is impossible for 3 generals with 1 traitor. ■

Simulation method works for $3f$ generals and f traitors. ■



BYZANTINE AGREEMENT ALGORITHMS

To reach agreement, processes have to exchange their values and replay the received values to other processes several times. The capability of faulty process to distort what it receive from the others greatly depends upon the type of messages.

Two types of messages:

- **oral messages** (non-authenticated) – a faulty process can forge a message and claim to have received it from another process or change the contents of a received message before it relays this message to other processes. There is no way for a process to verify the authenticity of a received message.
- **signed messages** (authenticated) – a faulty process cannot forge a message or change the contents of a received one before relaying it to other processes. Each process can verify the authenticity of a received message. Faulty processes do less damage.



System model

Let $\mathbb{S}$ be a system ($\mathbb{S}^{\text{Sync}}$ = synchronous, $\mathbb{S}^{\text{Async}}$ = asynchronous) as defined in lecture 1.

By $\mathbb{S}\{M\}$ we will denote system $\mathbb{S}$ enhanced with additional mechanism M .

- oral messages are available in $\mathbb{S}^{\text{Sync}}\{\emptyset\}$
- signed messages are available in $\mathbb{S}^{\text{Sync}}\{C\}$, where C is cryptographic digital signature



ORAL MESSAGES

This is a simple solution using communication by “oral” messages in $\mathbb{S}^{\text{Sync}}\{\emptyset\}$ to cope with f traitors, where $f < \frac{1}{3}N$

Assumptions:

Network is reliable =

A1: Every message sent is delivered correctly.

A2: The receiver of a message knows who sent it.

A3: The absence of a message can be detected.

A1 and A2 prevent a traitor from interfering with communication between others (A2 = no spurious messages).

A3 means that a traitor can't simply remain quiet to disallow progress.

Details:

More requirements:

- Generals communicate directly with each other (however this can be easily relaxed)
- If a message is missing, assume it says 0 (default order is RETREAT)

majority function:

- if more than $\frac{n}{2}$ of v_i equals v , then $majority(v_1, \dots, v_n) = v$
- if the majority of v_i does not exist: $majority(v_1, \dots, v_n) = 0$
- n is the number of processes currently reaching agreement



OM algorithm (Lamport-Shostak-Pease [1])

OM(0):

1. G_c sends his value to every G_i
2. Each lieutenant uses that value (or 0 if none)

OM(t), $t > 0$:

1. G_c sends his value to every G_i
2. Let v_i be the value received by G_i
 G_i now
 - acts as commander for OM($t-1$) with $n-2$ other lieutenants
 - acts as participant for $n-2$ instances of OM($t-1$), with each $G_j: i \neq j$ as commander, deciding v_j (or 0)
3. decide *majority*($v_1, \dots, v_{n-1}$)

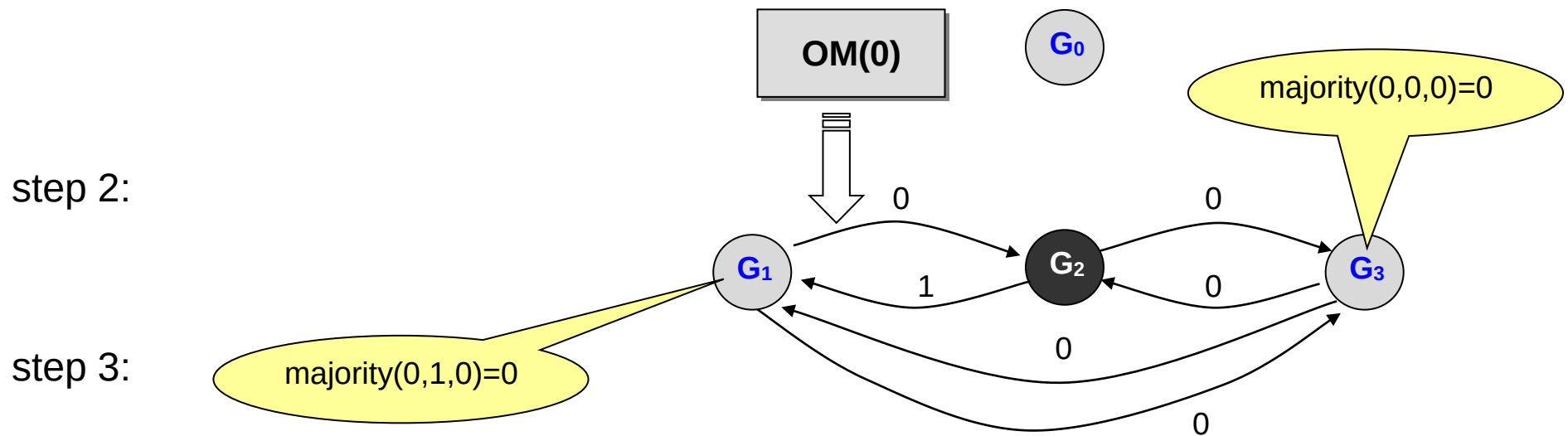
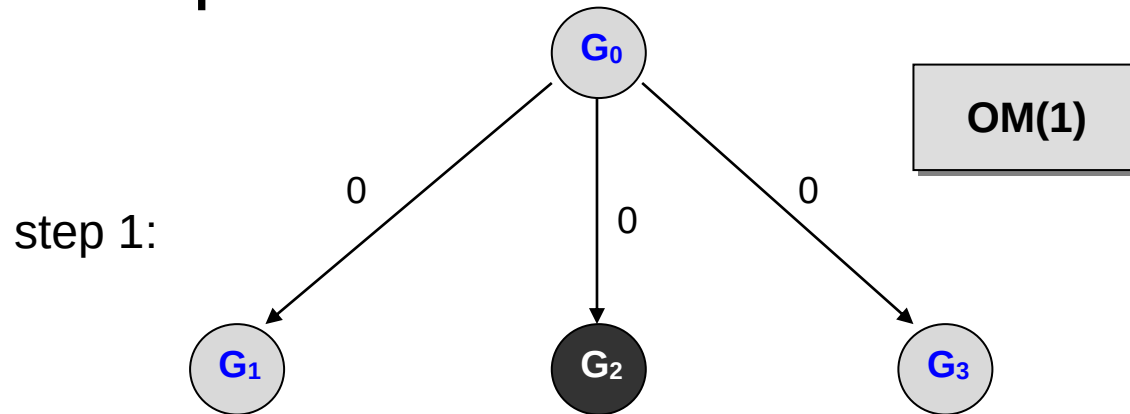


Description:

- algorithm for N processes starts with $OM(f)$
- execution of $OM(f)$ invokes $N-1$ separate executions of $OM(f-1)$, each of which invokes $N-2$ executions of $OM(f-2)$, and so on ...
- processes are successively divided into smaller and smaller groups of n members (initially $n=N$) and the Byzantine Agreement is recursively achieved within each group in step 2 of $OM(t)$.
- in total, there are $1(N-1)(N-2)(N-3)\dots(N-f)$ separate executions of $OM(t)$: $t = f, f-1, f-2, f-3, \dots, 0$, accordingly.
- the message complexity is $\Omega(N^{f+1})$
- algorithm requires $f+1$ rounds of message exchanges
- $f+1$ is the lowest bound on the number of rounds to reach Byzantine Agreement in a fully connected network with process failures in $\mathbb{S}^{\text{Sync}}\{\emptyset\}$

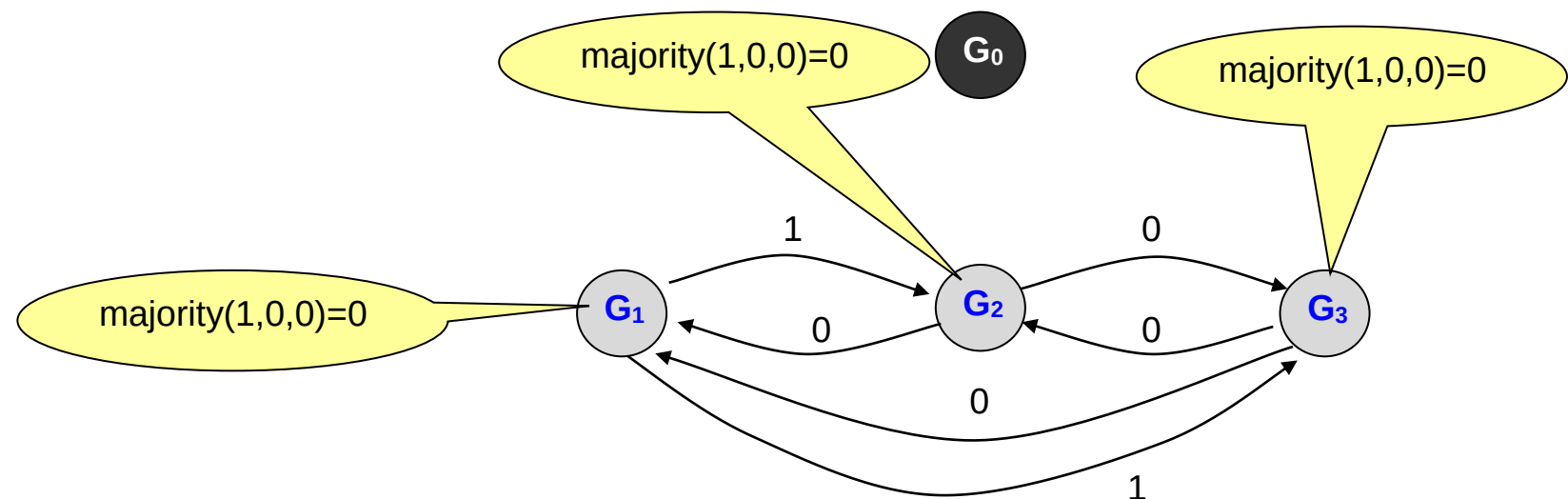
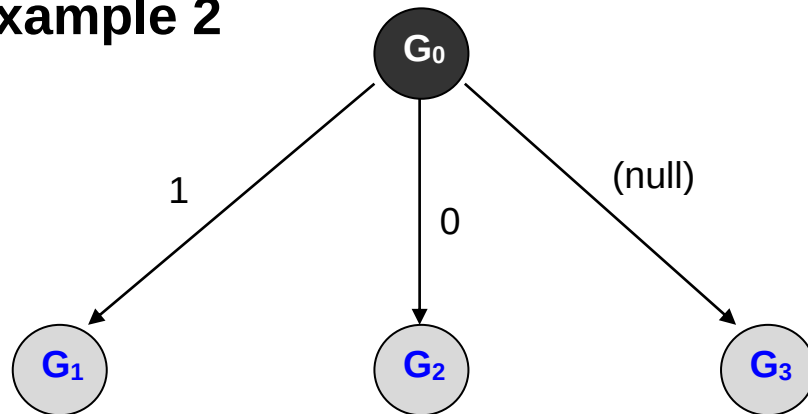


Example 1



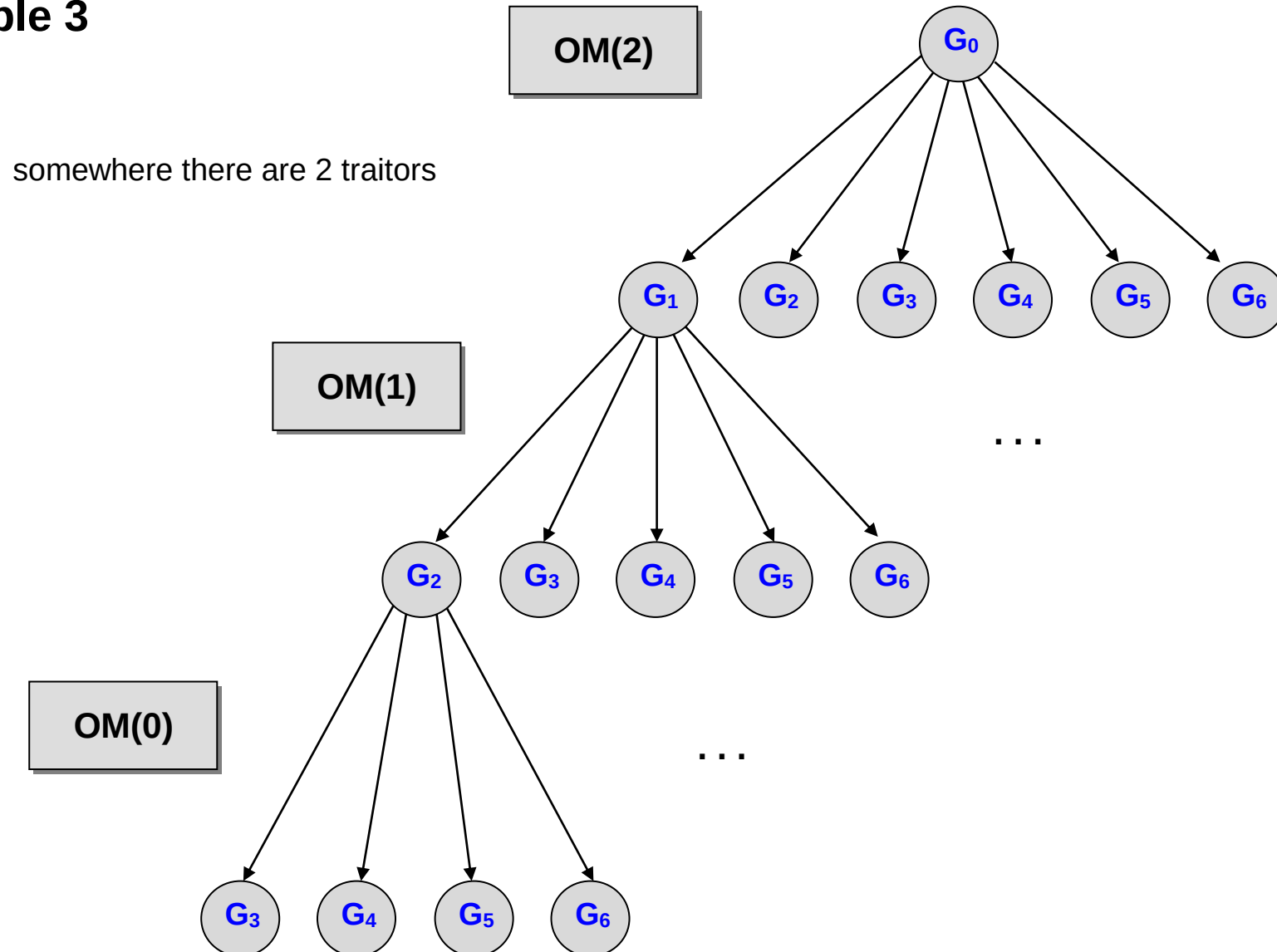


Example 2





Example 3





Correctness proof

As a disloyal commander causes more problems than any disloyal lieutenant, we consider these cases separately:

Lemma: For any t and f , OM(t) satisfies *Validity* if there are more than $2f + t$ generals and at most f traitors.

Recall:

Validity: If the commander is loyal, then every loyal lieutenant obeys the order he sends.

Lemma proof:

- we consider the case of a loyal commander
- OM($t=0$) trivially satisfies *Validity* (by assumption A1)
- proceed by induction for $t > 0$:
 - assume the lemma for OM($t-1$)
 - prove it for OM(t)



Induction step for OM(t), $t > 0$:

- loyal commander sends v to $n-1$ lieutenants
- each loyal lieutenant uses v invoking OM($t-1$) with $n-2$ other lieutenants
- by lemma hypothesis: $n > 2f + t$
- so $(n-1) > 2f + (t-1)$
- apply induction hypothesis: OM($t-1$) satisfies *Validity*
- when a loyal G_i is commander in OM($t-1$)
- each loyal lieutenant G_j gets the same v_i (from that loyal G_i) and uses it in OM(t)
- there are at most f traitors
- and $(n-1) > 2f + (t-1) \geq 2f$
- so majority of $n-1$ lieutenants in OM(t) are loyal
- each loyal lieutenant gets the same *majority*($v_1, \dots, v_{n-1}$) in step 3 of OM(t)





Theorem: For any f , $OM(f)$ satisfies *Agreement* and *Validity* if there are more than $3f$ generals and at most f traitors.

Theorem proof (by induction on f):

- if there are no traitors – $OM(0)$ – easy
- assume claim for $OM(f-1)$, $f > 0$
- assume commander is loyal
 - take $t = f$ in previous Lemma
 - $OM(f)$ satisfies *Validity* from Lemma
 - *Validity* implies *Agreement* if commander is loyal
- assume commander is a traitor...

- if commander is a traitor:
 - at most f traitors, including commander
 - so, at most $f-1$ traitors among the lieutenants
 - since $n > 3f$, lieutenants are more than $3f-1 > 3(f-1)$
 - apply **induction hypothesis**: OM($f-1$) satisfies *Agreement* and *Validity*
 - thus, for all i , any 2 loyal lieutenants G_j, G_k get the same decision v_i from OM($f-1$) at the end of step 2 of OM(f)
 - it follows from *Agreement* of OM($f-1$)
 - hence, any 2 loyal lieutenants get the same vector $v_1, \dots, v_{n-1}$ in step 3 of OM(f)
 - therefore, any 2 loyal lieutenants get the same *majority*($v_1, \dots, v_{n-1}$) from OM(f)





Ad-hoc protocol:

- each general G_i makes a decision v_i and broadcasts it
- generals reach agreement based on the total of collected opinions:
 - if most of generals independently decide **ATTACK**, then attack
 - if most of generals independently decide **RETREAT**, then retreat
 - if they are divided, then ... hmm ... well, uh ... hmm ... who cares?
- it may possibly require several rounds

Byzantine failures:

- let's trust the loyal generals
- what's the problem? ... traitors !
- if you receive v_j from G_j , you can't trust him
- you must be careful, because you don't know who to trust

traitors can
forge messages



SIGNED MESSAGES

- we have $\mathbb{S}^{\text{Sync}}\{C\}$, where C is cryptographic tool able to digitally sign messages:
 $v:i$ = value v signed by G_i

Assumptions:

- A1: Every message sent is delivered correctly.
- A2: The receiver of a message knows who sent it.
- A3: The absence of a message can be detected.
- A4: digital signature of G_i cannot be forged
(any alteration of the message can be detected)
and anyone can verify the authenticity of a signature

Can we obtain in $\mathbb{S}^{\text{Sync}}\{C\}$ better results than in $\mathbb{S}^{\text{Sync}}\{\emptyset\}$?



SM algorithm:

- G_0 makes a decision v and sends a signed order $v:0$ (round 0)
- each G_i adds his own signature to that order and sends it to the other lieutenants (round 1), who add their signature and send it to others (round 2), etc. ...
- the valid order (i.e. correctly signed) in round k is in form of $v:0:i_1: \dots :i_k$
- every G_i propagates only valid orders
- $f+1$ rounds of order propagation: from phase 0 to round f
- in the last round: from among $f+1$ signatures in $v:0:i_1: \dots :i_f$ at least one comes from a correct process

for set V_i of valid orders:

$$\text{choice}(V_i) = \begin{cases} v & \text{if } v \text{ is the only decision in } V_i, \text{ i.e. } V_i = \{v\} \\ 0 & \text{otherwise (RETREAT)} \end{cases}$$



For G_i :

$V_i = \emptyset$

round 0:

G_0 **broadcasts** $v:0$ to all processes;

receive $v:0$

if $v:0$ is valid then $V_i = V_i \cup \{v\}$

round $k = 1..f$:

while $v':0:j_1:\dots:j_{k-1}$ is a new value received in round $k-1$

send $v':0:j_1:\dots:j_{k-1}:i$ to every lieutenant except $G_{j_1}, \dots, G_{j_{k-1}}$

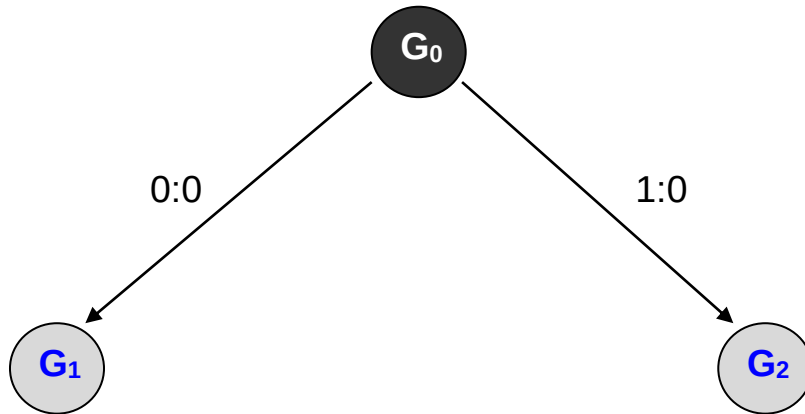
while receive $v:0:j_1:\dots:j_k$

if $v:0:j_1:\dots:j_k$ is valid then $V_i = V_i \cup \{v\}$

$v = \text{choice}(V_i)$



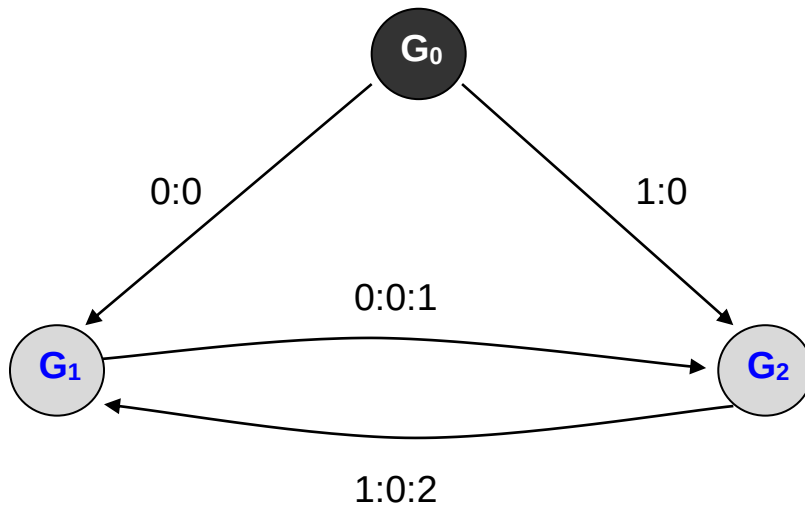
round 0:



- let $N=3$ and $f=1$
- with G_0 traitor



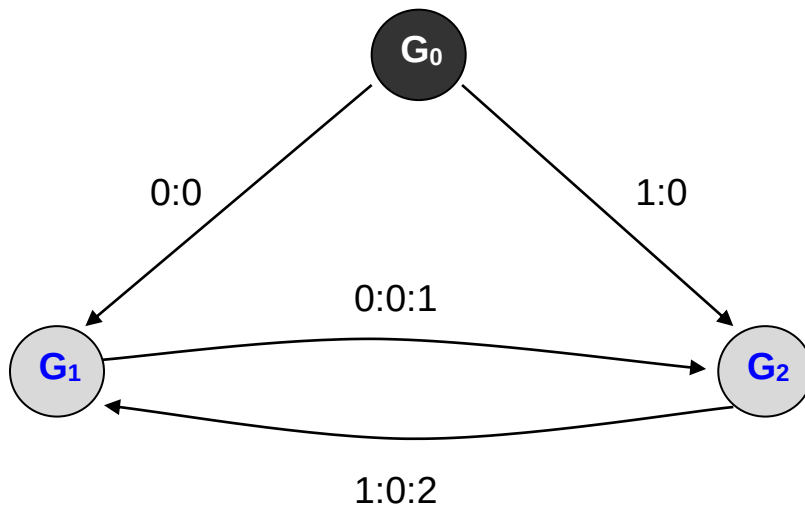
round 1:



- let $N=3$ and $f=1$
- with G_0 traitor
- G_1 and G_2 know that G_0 is a traitor
from A4 it follows that only he could
give conflicting orders



How many f ?



- let $N=3$ and $f=1$
- with G_0 traitor
- G_1 and G_2 know that G_0 is a traitor
from A4 it follows that only he could
give conflicting orders

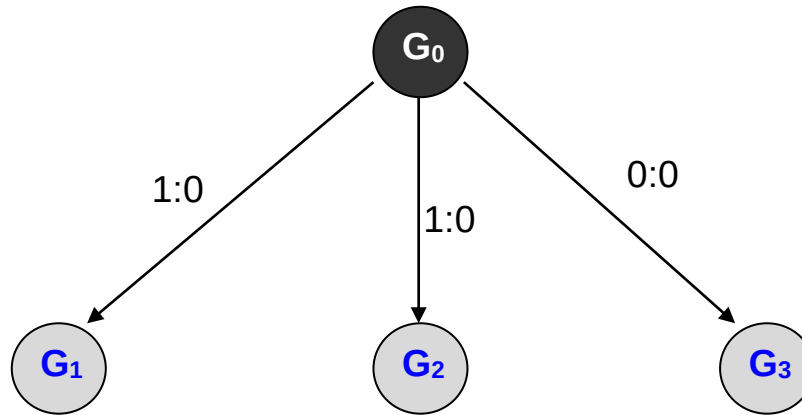
- formally any $f < N$
- obviously it is reasonable only for $N \geq f + 2$

common decision



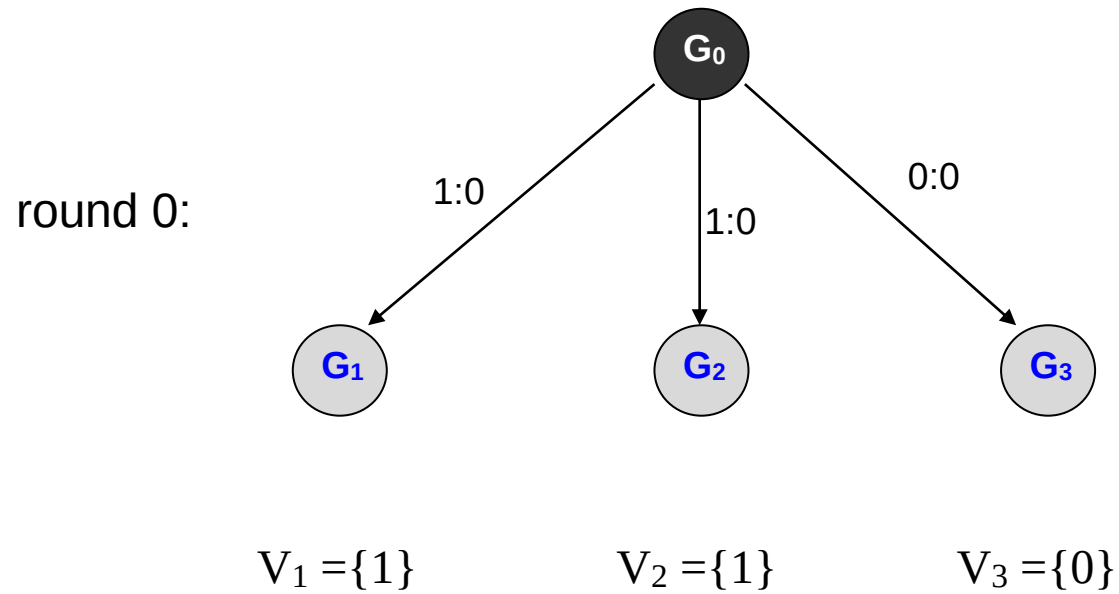
Really $f+1$ rounds?

round 0:



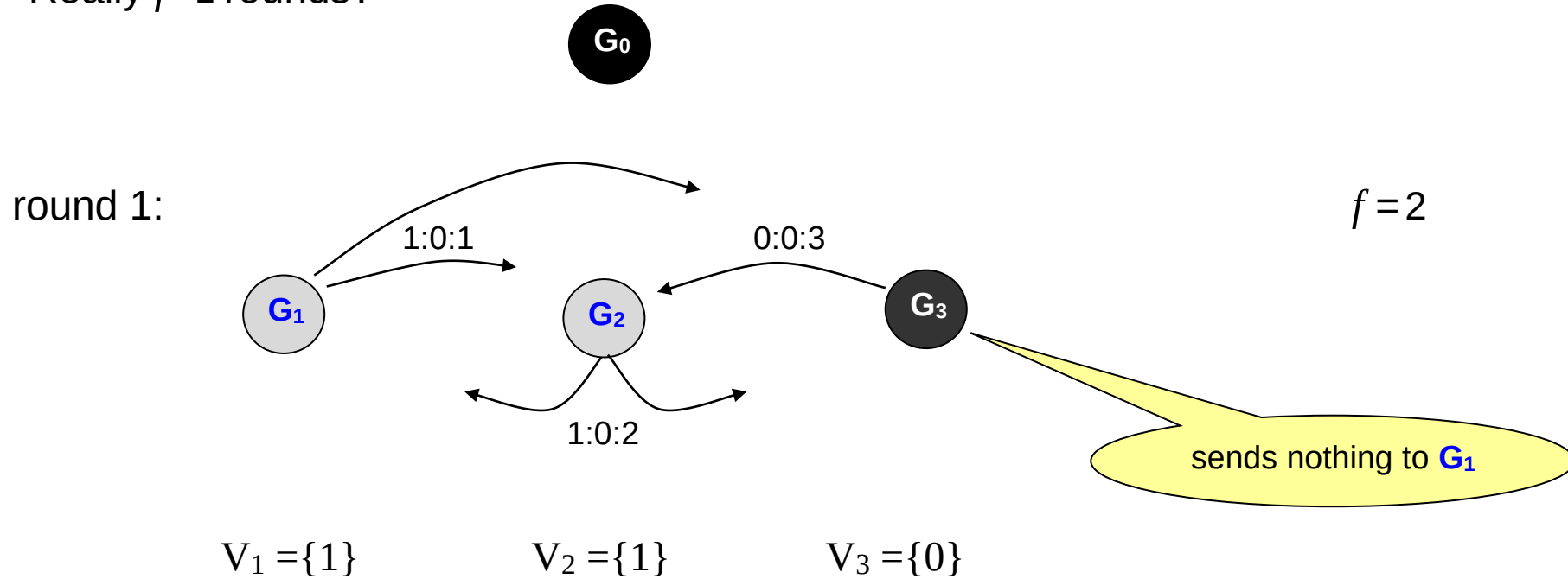


Really $f+1$ rounds?





Really $f+1$ rounds?

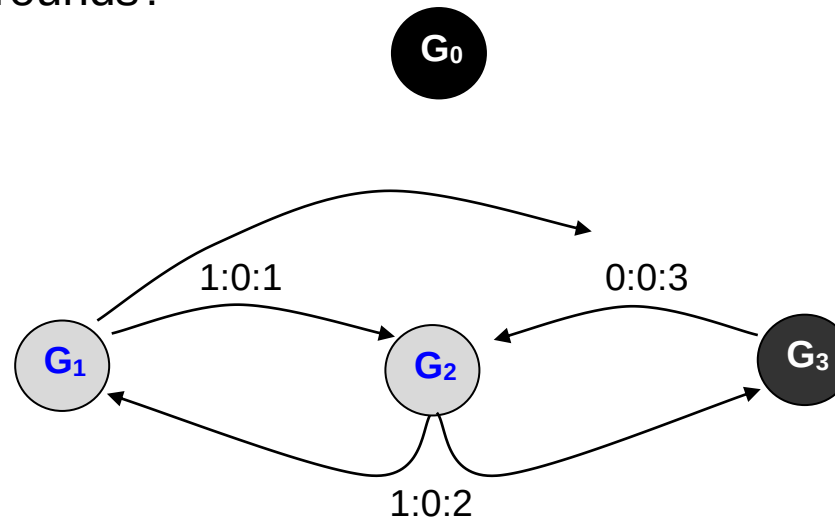


Here, if a message is missing, don't assume it says 0! Only signed orders matter!



Really $f+1$ rounds?

round 1:



$f=2$

$V_1 = \{1\}$

$V_2 = \{1,0\}$

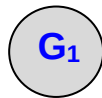


Really $f+1$ rounds?



round 2:

$f=2$



0:0:3:2

$V_1 = \{1\}$

$V_2 = \{1,0\}$



Really $f+1$ rounds?



round 2:

$f=2$



0:0:3:2

$V_1 = \{1,0\}$

$V_2 = \{1,0\}$

$\text{choice}(V_i) = 0$



Message complexity:

- $f+1$ rounds ($f < N$)
- $O(N^2)$:
 - round 0: $(N-1)$ messages
 - round 1: $(N-1)(N-1)$ messages
 - round 2: at most $(N-1)(N-2)$ messages
 - ...
 - round f : at most $(N-1)(N-f)$ messages
 - in total: at most $(N-1) + f(N-1)^2$ messages $\rightarrow O(N^2)$

Optimization:

G_i forwards only the first two valid messages that carries different values

- if G_0 is correct, he has sent the same signed value v to all the processes that can only forward it (after having appended their signatures) – this is because no process can forge the signature of another process – hence, a single valid value v can only be received by any correct G_i
- if G_0 is faulty, he can have sent different signed values to different processes and no message to other processes – if G_i receives several valid messages carrying different values, G_i knows that these values originated from the commander G_0 , and G_0 is faulty. To make other correct processes aware of that, G_i needs to forward (after having signed them) only two messages that carry different values



Message complexity after optimization:

- still $f+1$ rounds ($f < N$)
- and still $O(N^2)$ but:
 - round 0: $(N-1)$ messages
 - round 1: $(N-1)(N-1)$ messages
 - round 2.. f : at most $(N-1)(N-?)$ messages
 - in total: at most $(N-1) + 2(N-1)^2$ messages – if the commander is faulty
 - and at most $(N-1) + (N-1)^2$ messages if the commander is correct

BIBLIOGRAPHY

- [1]  L. Lamport, R. Shostak, M. Pease, "The Byzantine Generals Problem", *ACM Transactions on Programming Languages and Systems*, 1982.
- [2]  Michel Raynal, " Fault-Tolerant Message-Passing Distributed Systems ", Springer, 2018, ch. 14
- [3]  Ajay D. Kshemkalyani, Mukesh Singhal, "Distributed Computing – Principles, Algorithms, and Systems", Cambridge University Press, 2008, ch. 14